

Small models. Tight loops.

What a coding-agent harness is actually doing for you

one Python file

real compiler

external postconditions

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```
nanagent / run 01
```

```
$ askme "build a two-file C program"
```

```
plan   write .h → write .c → compile → run
```

```
tool   cc -o main main.c
```

```
check  ./main
```

AGENT_OK

The bug that ate minutes

```
// missing: #include <stdio.h>
int main() {
    printf("FIXED\n");
    return 0;
}

error: implicit declaration of printf
```

step 1

Compile

Useful failure: the compiler names the bug.

next

Bad edit

Malformed JSON or the wrong exact-match string.

140–253s

Think, retry, re-read

The model spends tokens rediscovering known state.

Wrong move: buy more thinking. **Better move:** turn the compiler error into control flow.

Move reliability out of the model



```
compiler stderr → [compile_error] → repair / retry → ./main == "AGENT_OK"
```

Six runs, two concrete jobs

MULTI-FILE BUILD

Header + source → compile → run

```
msg.h + main.c · ./main == REPLAN_OK
```

REPAIR

Fix a Python syntax error

```
python3 greet.py exits 0 · stdout contains hello
```

Gemma 4 26B A4B

MoE · 3.8B active

Qwen3.6-27B

dense · 27B

Qwen3.6-35B-A3B

MoE · 3B active

Held fixed: NanAgent commit, prompts, SiliconFlow routing with FP8 endpoints, reasoning off initially with one retry policy, strict external checks. **n=1 per cell.**

DRAFT RESULT · REPLACE AFTER AUTHENTICATED RUN

What happened?

Model	Build + run	Repair + run	LLM calls	Billed credits
Gemma 4 26B A4B	PENDING	PENDING	—	—
Qwen3.6-27B	PENDING	PENDING	—	—
Qwen3.6-35B-A3B	PENDING	PENDING	—	—

Pass means both **agent complete** and the deterministic postcondition passed. Latency is shown per task because it is provider- and network-dependent.

Read this as a smoke test. One run can find an integration failure; it cannot estimate a success rate.

Useful evidence, small claim

What we actually verify

- Structured actions survive the model/provider path
- The loop can build or repair these exact programs
- The resulting artifact executes correctly
- Calls, tokens, route, and cost are auditable

What we do not claim

- Full-app or long-horizon reliability
- Local-laptop speed from hosted runs
- UX, architecture, or maintainability quality
- Contest-code scores predict agent performance

TAKEAWAY

The loop is the product

Model

Typed action

Guardrail

Real tool

Postcondition

**Small models do not need easier standards.
They need tighter feedback loops.**

github.com/den-run-ai/askme · slides, blog, protocol, and raw summary data